module6_lab3

October 11, 2026

[1]:
```
#Step 1 rebuild Module 6 Lab 2 Tables
```

[2]:
```
import sqlite3
import pandas as pd
```

[3]:
```
con = sqlite3.connect("world.db")
```

[4]:
```
con.execute("DROP TABLE IF EXISTS City;")
```

[4]:
```
<sqlite3.Cursor at 0x190c6d72ec0>
```

[5]:
```
con.execute("""
CREATE TABLE 'City' (
'ID' INTEGER PRIMARY KEY AUTOINCREMENT,
'Name' char(35) NOT NULL default '',
'CountryCode' char(3) NOT NULL default'',
'District' char(20) NOT NULL default'',
'Population' int(11) NOT NULL default '0'
);""")
```

[5]:
```
<sqlite3.Cursor at 0x190c6d72640>
```

[6]:
```
con.execute("INSERT INTO City VALUES (NULL, 'New York', 'USA', 'New York',␣
 ↪10000000);")
con.execute("INSERT INTO City VALUES (NULL, 'Los Angeles', 'USA', 'California',␣
 ↪4000000);")
con.execute("INSERT INTO City VALUES (NULL, 'Orlando', 'USA', 'Florida',␣
 ↪2700000);")
con.execute("INSERT INTO City VALUES (NULL, 'Dallas', 'USA', 'Texas', 2300000);
 ↪")
con.execute("INSERT INTO City VALUES (NULL, 'Las Vegas', 'USA', 'Nevada',␣
 ↪1600000);")
con.execute("INSERT INTO City VALUES (NULL, 'Columbus', 'USA', 'Ohio', 1500000);
 ↪")
con.execute("INSERT INTO City VALUES (NULL, 'Atlanta', 'USA', 'Georgia',␣
 ↪1400000);")
```

```
con.execute("INSERT INTO City VALUES (NULL, 'Pittsburgh', 'USA',␣
 ↪'Pennslyvania', 1400000);")
con.execute("INSERT INTO City VALUES (NULL, 'Salt Lake City', 'USA', 'Utah',␣
 ↪1200000);")
con.commit()
```

[7]:
```
df_city = pd.read_sql("SELECT * FROM City", con)
display(df_city)
```

	ID	Name	CountryCode	District	Population
0	1	New York	USA	New York	10000000
1	2	Los Angeles	USA	California	4000000
2	3	Orlando	USA	Florida	2700000
3	4	Dallas	USA	Texas	2300000
4	5	Las Vegas	USA	Nevada	1600000
5	6	Columbus	USA	Ohio	1500000
6	7	Atlanta	USA	Georgia	1400000
7	8	Pittsburgh	USA	Pennslyvania	1400000
8	9	Salt Lake City	USA	Utah	1200000

[8]:
```
# Step 2 Using a WHERE - Population is < 2,300,000
```

[9]:
```
query_where="SELECT Name, Population FROM City WHERE Population < 2300000;"
df_where = pd.read_sql(query_where,con)
display(df_where)
```

	Name	Population
0	Las Vegas	1600000
1	Columbus	1500000
2	Atlanta	1400000
3	Pittsburgh	1400000
4	Salt Lake City	1200000

[10]:
```
# Step 3 Using an ORDER - Alphabetically
```

[11]:
```
query_order="SELECT Name, District FROM City ORDER BY District ASC;"
df_order = pd.read_sql(query_order,con)
display(df_order)
```

	Name	District
0	Los Angeles	California
1	Orlando	Florida
2	Atlanta	Georgia
3	Las Vegas	Nevada
4	New York	New York
5	Columbus	Ohio
6	Pittsburgh	Pennslyvania
7	Dallas	Texas
8	Salt Lake City	Utah

[12]:
```
# Step 4 Using a LIKE - Finding Names with -as- in it
```

[13]:
```
query_like="SELECT Name, District, Population FROM City WHERE Name LIKE '%as%';"
df_like = pd.read_sql(query_like,con)
display(df_like)
```

	Name	District	Population
0	Dallas	Texas	2300000
1	Las Vegas	Nevada	1600000

[14]:
```
# Step 5 Chart of Population
```

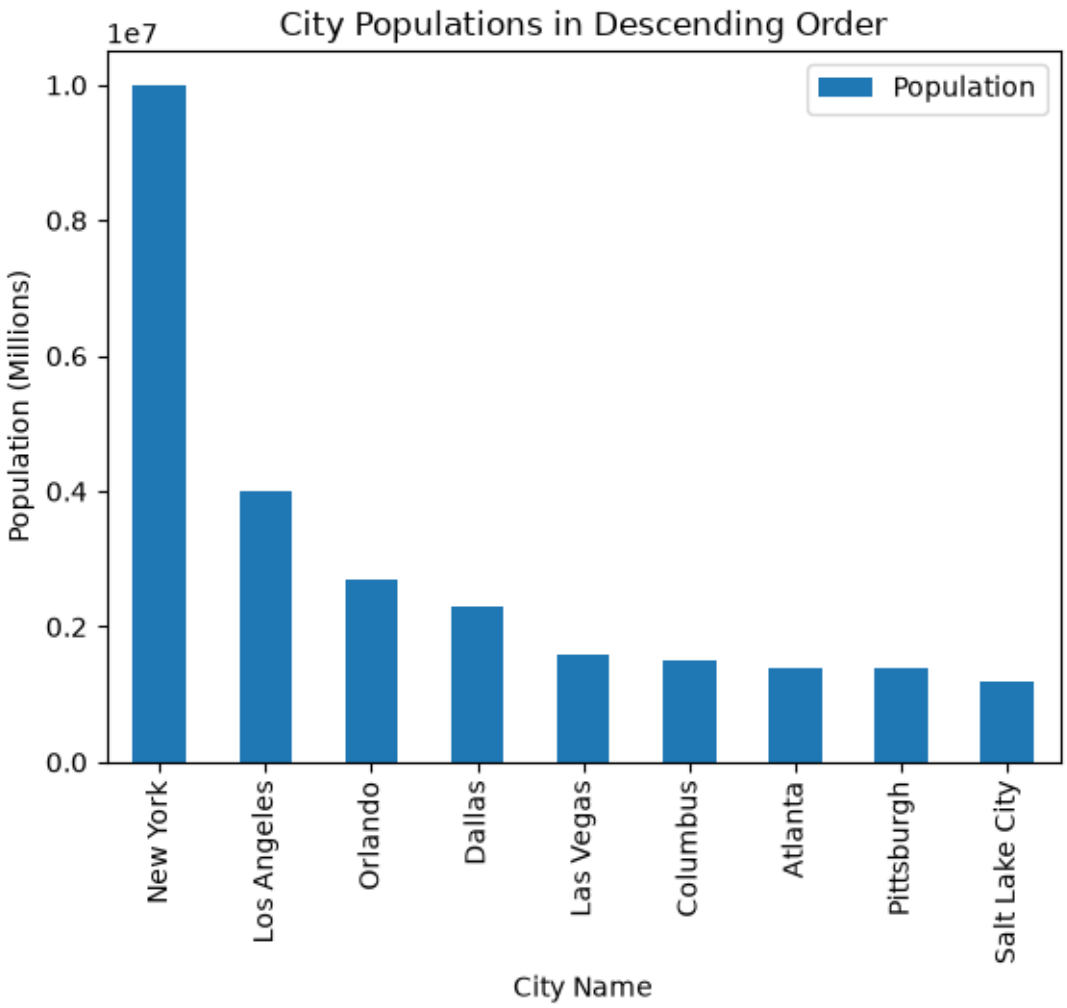
[15]:
```
import matplotlib.pyplot as plt
```

[16]:
```
df_chart = pd.read_sql("SELECT Name, Population FROM City ORDER BY Population␣
↪DESC;", con)
```

[17]:
```
df_chart.plot(
    kind="bar",
    x="Name",
    y="Population")

plt.title("City Populations in Descending Order")
plt.xlabel("City Name")
plt.ylabel("Population (Millions)")
```

[17]:
```
Text(0, 0.5, 'Population (Millions)')
```


City Populations in Descending Order
1e7
Population
1.0
0.8
0.6
0.4
0.2
0.0
Population (Millions)
New York
Los Angeles
Orlando
Dallas
Las Vegas
Columbus
Atlanta
Pittsburgh
Salt Lake City
City Name

[ ]: